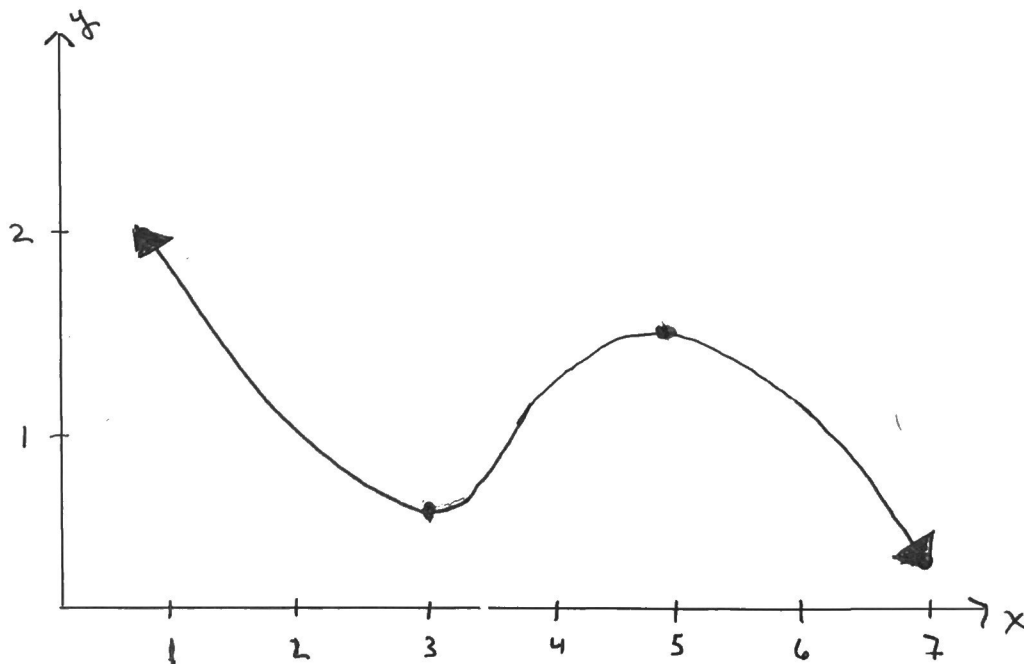


Unit 3 – Curve Sketching

Increasing & Decreasing Functions

- A function is **increasing** on an interval if the graph of the function **increases** from **left to right**.
- A function is **decreasing** on an interval if the graph of the function **decreases** from **left to right**.

Example: Locate the intervals of increase and decrease for the function below.



Increasing: $(3, 5)$

Decreasing: $(-\infty, 3) \cup (5, \infty)$

- A function is **increasing** on an interval if the function has a **positive** slope of the tangent on that interval.
- A function is **decreasing** on an interval if the function has a **negative** slope of the tangent on that interval.

Test for Increasing and Decreasing Functions

- If $f'(x) > 0$ for all x in an interval, then $f(x)$ is **increasing** in that interval.
- If $f'(x) < 0$ for all x in an interval, then $f(x)$ is **decreasing** in that interval.

Critical Numbers

- A **critical number**, c , is a number in the domain of $f(x)$ such that

$$f'(c) = 0 \quad \text{OR} \quad f'(c) \text{ DNE.}$$

Example: Determine the intervals on which the following functions are increasing or decreasing.

a) $f(x) = 3x^2 + 6x - 1$

$$f'(x) = 6x + 6$$

$$\therefore \text{critical \#s} \Rightarrow f'(x) = 0$$

$$6(x+1) = 0$$

$$x = -1 \quad (\text{critical \#})$$

since there is 1 critical # $\Rightarrow$ 2 intervals.

Interval	$(-\infty, -1)$	$(-1, \infty)$
$x+1$	-	+
$f'(x)$	-	+
$f(x)$	$\downarrow$	$\uparrow$
	decreasing D	increasing I

$\therefore$ Decreasing $(-\infty, -1)$

Increasing $(-1, \infty)$

b) $f(x) = x^3 + 9x^2 + 24x - 3$

$f'(x) = 3x^2 + 18x + 24$

$= 3(x^2 + 6x + 8)$

$\therefore$ critical #'s $\Rightarrow f'(x) = 0$

$3(x^2 + 6x + 8) = 0$

$3(x+4)(x+2) = 0$

$\therefore x = -4, -2 \Rightarrow 3 \text{ intervals}$

Interval	$(-\infty, -4)$	$(-4, -2)$	$(-2, \infty)$
$x+4$	-	+	+
$x+2$	-	-	+
$f'(x)$	+	-	+
$f(x)$	↑	↓	↑
	increasing I	decreasing D	increasing. I

$\therefore$ Increasing: $(-\infty, -4) \cup (-2, \infty)$

Decreasing: $(-4, -2)$

$$c) f(x) = \frac{1}{4}x^4 + \frac{4}{3}x^3 - \frac{5}{2}x^2$$

$$f'(x) = x^3 + 4x^2 - 5x$$

$$= x(x^2 + 4x - 5)$$

$$= x(x+5)(x-1)$$

critical #'s $\Rightarrow f'(x) = 0$

$\therefore x = 0, -5, 1 \Rightarrow 4$ intervals

Interval	$(-\infty, -5)$	$(-5, 0)$	$(0, 1)$	$(1, \infty)$
x	-	-	+	+
$x+5$	-	+	+	+
$x-1$	-	-	-	+
$f'(x)$	+	+	-	+
$f(x)$	↓ D	↑ I	↓ D	↑ I

Increasing $(-5, 0) \cup (1, \infty)$

Decreasing $(-\infty, -5) \cup (0, 1)$